

Autosave

On

PROG-SPEC_219870_LOTT1_refresh_W0.1 - Onkar.xlsm

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fx

Instruction: Delete any examples in grey italic text that are not relevant/applicable to the study once the programming specification has been finalized

Section A: Data Source Selection

Data source	Population Subset	Version
Optum	Clinformatics Data Mart (CDM)	2025q2

Section B: Define Study Time Periods

Study Time Period	Label	Definition	Additional Notes	Date Modified (Initials + Date)	QC Reviewed (Initials + Date)
STUDYPD	Full study period	01Jul2015 - 30June2025 (inclusive)	Full study time period		OK19012026
ID_PD	Patient identification period	01Jan2016 - 30June2025 (inclusive)	Allows for 6mos lookback and no required minimum length of follow-up. ID_PD end at ends at the earliest of death, or the end of the study period For CE, 3mosF=1 follow-up ends at earliest of death, health plan disenrollment or study period end		OK19012026
INDEX	Index date	Date of the patient's earliest qualifying multiple myeloma diagnosis date	Patients are required to have at least 1 inpatient cancer visit for multiple myeloma (ICD-9-CM=203.0x or ICD-10-CM code=C90.0x) or at least 2 outpatient visits in any position on claim, on separate days within 90 days, during identification period. (Patients with 2 claims 30 or 60 days apart will also be flagged for consideration.) The index date is the date of the first medical claim for multiple myeloma. Diagnosis codes for MM are in tab 42 "MM dx code". Flags should also be made to distinguish which claims are non-diagnostic claims. A non-diagnostic claim refers to a claim for treatment of a condition (e.g., infusion) rather than diagnosis (see tab 49 "Diagnostic"). Patients must have six months of CE with medical or pharmacy benefits before the index date. Patients with gaps in enrollment of ≤ 30 days are considered to be continuously enrolled		OK19012026
Baseline	Baseline period	The baseline period is fixed and is the six months before the index date	Patients are only required to have ≥ 1 day of CE starting on the index date. Patients with gaps in enrollment of ≤ 30 days are considered to be continuously enrolled.		OK19012026

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3.Data Prep

4.StudyPop

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5B.MAP_MED

6.LOTT_BASE

7.SCT

10.LOTT_BASE_END OK

40.CL MMA ROLLUP

41.CL MMA CODELIST

41a.MMA NDC CODELIST

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A	B	C	D	E	F	G
	FU_PD	Follow up period	The follow up period is fixed and is from the index date till the end of the patient identification period (Inclusive).	Patients are only required to have ≥ 1 day of CE starting on the index date. Patients with gaps in enrollment of ≤ 30 days are considered to be continuously enrolled, consider medical and/or pharmacy benefits. Follow up period ends the earliest of death or the end of the study period (30 June 2025).		OK19012026
	e.g. PST24 etc.	e.g. 24-month post-index period	e.g. INDEX_DT through INDEX_DT + 730			

Section C: Source Data Preparation

Instruction: Section C should ONLY be used to document decisions to the source data before the analytic file build. If decisions apply to a variable in the analytic file, document in the appropriate variable definition.

#	Source Table/Variable	Situation Requiring Resolution	Action	Reasoning / Comments	Date Modified (Initials + Date)	QC Reviewed (Initials + Date)
1	Days to months conversion	Assuming 30.4375 day per month	3 months = 91 days, 6 months=183 days, 9 months=274 days, 12 months=365 days	The team will additionally include patients who had ≥ 1 non-diagnostic medical claim for MM during the baseline period. These patients were excluded by Optum originally if a patient has newly diagnosed or active MM. It would be likely that they would experience frequent activity, hence why we look for 2 non-diagnostic MM claims close together in the first step of the attrition table. A smoldering patient would have infrequent activity, hence why a claim for MM that falls outside of the 30, 60 or 90 days window requirement that's used to identify newly diagnosed patients is likely a smoldering patient.		
2	ELIG_COH	Study team does not want to exclude smoldering MM patients. Optum built study cohort with the intention to exclude smoldering MM patients.		Study team will consider excluding patients with a stand alone treatment code (rather than any non-diagnostic code) on an MM claim in the baseline period for future sensitivity analysis. This would not remove smoldering patients but rather ensure our patients were newly active MM patients.		OK19012026
3	FU_PD	Original IE criteria from Optum only considers patients with at least 6 mos of CE from index onwards.	Study team wants to keep criteria as broad as possible.	Will only require at least a day of CE post index, though will flag patients with 3 or more months of CE, to potentially subset on these patients for later consideration.		OK19012026
4	MED_COND	Conditioning medications are included in MM-	Ignore these treatments for now.	The team is not particularly interested in conditioning therapy. These treatments potentially will be added		OK19012026

Approach. We implement both Approach 1 and Approach 2 in combination (OR logic) per Optum CDM business rules guidance.

Approach 1 (POSTOS): Inpatient if POS IN (21, 51, 61) OR TOS_CD IN ('FAC_IP_ACUTE', 'FAC_IP_REHSNF', 'PROF_INPVS', 'FAC_IP_SNF')

Approach 2 (CONF_ID): Inpatient if CONF_ID is NOT NULL and validated against T_CONFINEMENT table with both ADMIT_DATE and DISCH_DATE present

Non-inpatient: All records not meeting either approach above

A claim is classified as inpatient if either approach identifies it as inpatient. This maximizes capture of true inpatient encounters.

Rationale: Using both approaches together provides the most comprehensive inpatient identification. From our validation on the 2025q2 data:

Total patients (Step 0): 94951
Any inpatient event: 43439 (45.2%)
Via POSTOS (Appr 1): 42618 (44.3%)
Via CONF_ID (Appr 2): 40357 (42.0%)
Both approaches: 39536 (41.1%)
POSTOS only: 3082 (3.2%)
CONF_ID only: 821 (0.9%)
Outpatient only: 52757 (54.8%)

OK19012026

For this LOT algorithm, the team has decided not to define maintenance therapy as its own LOT but rather to flag the inclusion of LOT1 regimens that include potential valid maintenance therapies/therapy combinations.

Reporting mtx is not of importance/relevance to upcoming studies planned in Multiple Myeloma. Flags will enable users to define mtx themselves in the future if so needed.